



Fig. 4: Author's prototype of the touch switch panel

The touch detector circuit has an oscillator whose frequency is dependent on capacitance of the touchpad. When a finger is moved close to the touchpad, additional capacitance causes frequency of this internal oscillator to change. The detector circuit tracks oscillator frequency at timed intervals, and when the shift crosses the threshold change, the circuit triggers a key-press event.

Touch switch detector

TTP223 is a one-touch keypad detector IC, while TTP224 and TTP225 can manage four and eight pads, respectively. The block diagram of TTP223 IC is shown in Fig. 2.

TTP223 IC is available in different packages having sixteen-, eight- or six-pin configurations to suit different applications. In this project, six-pin version is used, and its pin-out is shown in Fig. 3(a).

Behaviour of this package can be altered by making some changes on its pins 4 and 6.

Pin 4(AHLB). Output is active high when this pin is open, and active low when it is connected to VDD (positive rail).

Pin 6(TOG). This pin determines the mode of operation. If it is open, output remains in active state as long as the touchpad is touched. If this pin is high(VDD), it operates in toggle mode.

Prebuilt modules built around this single touchpad detector IC TTP223 have been used in this project. These modules are inexpensive and easily available on online portals. The circuit diagram and PCB module are shown

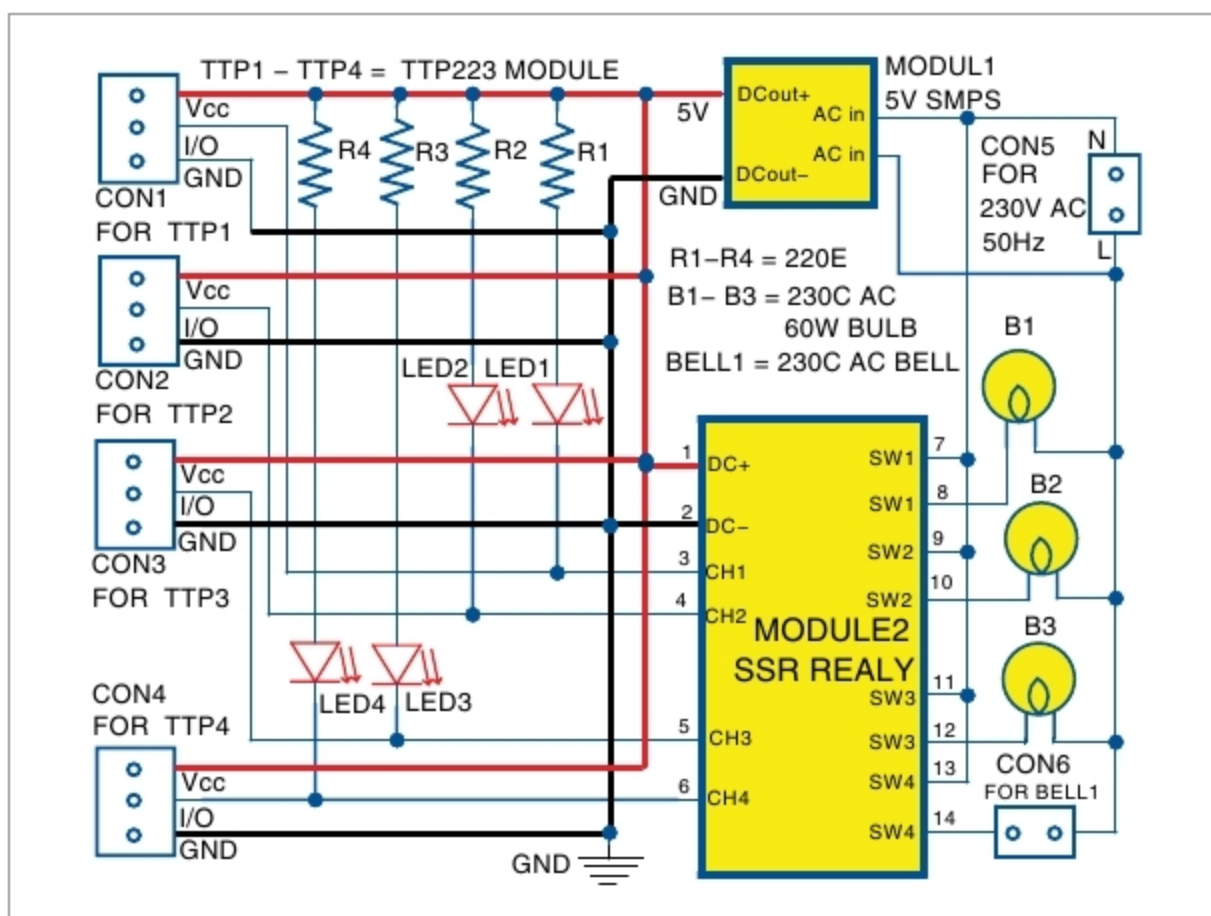


Fig. 5: Circuit diagram of the four-channel touch switch

in Figs 3(a) and 3(b), respectively.

Circuit and working

For this touch switch panel, four touch switches are used. Three switches function like normal on/off switches, while the fourth works as a momentary switch (bell switch). The number of switches on the panel can be varied depending on requirement.

In this project, a suitable-size Bakelite switch box is used to enclose the circuit. Take an acrylic sheet and cut it to the size of the box to make the front touch panel. Various input labels are marked on the acrylic front panel. The author's prototype of the touch switch panel is shown in Fig. 4.

Before using off-the-shelf touch switch modules, we need to modify them to change their default behaviour. For light switches, two solder pads, marked A and B on the module, have to be shorted. For the bell switch, only pad A is to be shorted, whereas pad B is to be left open. Pad A has to be shorted to match the relays, which have active low inputs.

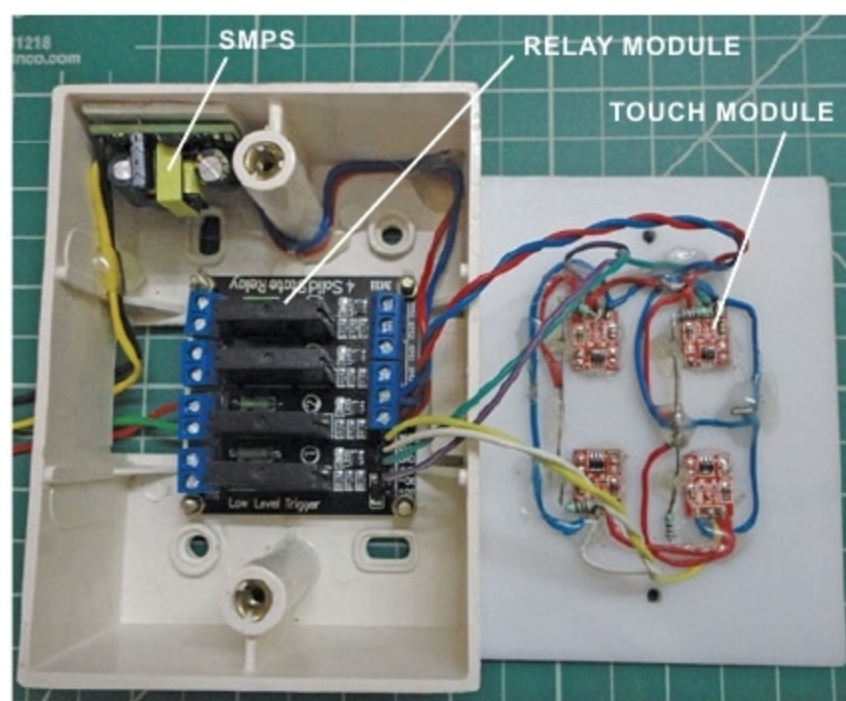


Fig. 6: Internal portion of the prototype shown in Fig. 4

Further, there is an LED indicator on the module that glows when output is high, as shown in Fig. 3(a). Since this relay module requires an active low signal, indicator LED on the module is always glowing in off state. Remove the indicator LED from each module and instead connect an external 3mm LED, as shown in Fig. 5. Glowing of LED will indicate on state of the appliance on the front panel.

The panel uses four touch switch modules, a four-relay module and a 5V SMPS module to power the circuit. Though any type of relay board can be used, a solid-state relay (SSR) module and a 700mA, 5V SMPS power